

Foundations of LLaMA language model

[LLaMA language model]

1.0 Content Block

Leak On March 3, 2023, a torrent containing Llama's weights was uploaded, with a link to the torrent shared on the 4chan imageboard and subsequently spread through online AI communities. That same day, a pull request on the main Llama repository was opened, requesting to add the magnet link to the official documentation. On March 4, a pull request was opened to add links to HuggingFace repositories containing the model. On March 6, Meta filed takedown requests to remove the HuggingFace repositories linked in the pull request, characterizing it as "unauthorized distribution" of the model. HuggingFace complied with the requests. On March 20, Meta filed a DMCA takedown request for copyright infringement against a repository containing a script that downloaded Llama from a mirror, and GitHub complied the next day. Reactions to the leak varied. Some speculated that the model would be used for malicious purposes, such as more sophisticated spam. Some have celebrated the model's accessibility, as well as the fact that smaller versions of the model can be run relatively cheaply, suggesting that this will promote the flourishing of additional research developments. Multiple commentators, such as Simon Willison, compared Llama to Stable Diffusion, a text-to-image model which, unlike comparably sophisticated models which preceded it, was openly distributed, leading to a rapid proliferation of associated tools, techniques, and software.

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Llama 1 models are only available as foundational models with self-supervised learning and without fine-tuning. Llama 2 – Chat models were derived from foundational Llama 2 models. Unlike GPT-4 which increased context length during fine-tuning, Llama 2 and Code Llama - Chat have the same context length of 4K tokens. Supervised fine-tuning used an autoregressive loss function with token loss on user prompts zeroed out. The batch size was 64. For AI alignment, human annotators wrote prompts and then compared two model outputs (a binary protocol), giving confidence levels and separate safety labels with veto power. Two separate reward models were trained from these preferences for safety and helpfulness using reinforcement learning from human feedback (RLHF). A major technical contribution is the departure from the exclusive use of proximal policy optimization (PPO) for RLHF – a new technique based on rejection sampling was used, followed by PPO. Multi-turn consistency in dialogs was targeted for improvement, to make sure that "system messages" (initial instructions, such as "speak in French" and "act like Napoleon") are respected during the dialog. This was accomplished using the new "Ghost attention" technique during training, which concatenates relevant instructions to each new user message but zeros out the loss function for tokens in the prompt (earlier parts of the dialog).

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Llama 3 On April 18, 2024, Meta released Llama 3 with two sizes: 8B and 70B parameters. The models have been pre-trained on approximately 15 trillion tokens of text gathered from "publicly available sources" with the instruct models fine-tuned on "publicly available instruction datasets, as well as over 10M human-annotated examples". Meta AI's testing showed in April 2024 that Llama 3 70B was beating Gemini Pro 1.5 and Claude 3 Sonnet on most benchmarks. Meta also announced plans to make Llama 3 multilingual and multimodal, better at coding and reasoning, and to increase its context window. Regarding scaling laws, Llama 3 models empirically showed that when a model is trained on data that is more than the "Chinchilla-optimal" amount, the performance continues to scale log-linearly. For example, the Chinchilla-optimal dataset for Llama 3 8B is 200 billion tokens, but performance continued to scale log-linearly to the 75-times larger dataset of 15 trillion tokens. During an interview with Dwarkesh Patel,

Mark Zuckerberg said that the 8B version of Llama 3 was nearly as powerful as the largest Llama 2. Compared to previous models, Zuckerberg stated the team was surprised that the 70B model was still learning even at the end of the 15T tokens training. The decision was made to end training to focus GPU power elsewhere. Llama 3.1 was released on July 23, 2024, with three sizes: 8B, 70B, and 405B parameters.

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Training datasets Llama's developers focused their effort on scaling the model's performance by increasing the volume of training data, rather than the number of parameters, reasoning that the dominating cost for LLMs is from doing inference on the trained model rather than the computational cost of the training process. Llama 1 foundational models were trained on a data set with 1.4 trillion tokens, drawn from publicly available data sources, including:

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The Llama 4 series was released in 2025. The architecture was changed to a mixture of experts where only a fraction of the model's expert sub-networks are activated per input token. They are multimodal (text and image input, text output) and multilingual (12 languages). Specifically, on 5 April 2025, the following were released both as base and instruction-tuned versions:

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Initial release The first version of Llama (stylized as LLaMA and sometimes referred to as Llama 1) was announced on February 24, 2023, via a blog post and a paper describing the model's training, architecture, and performance. The inference code used to run the model was publicly released under the open-source GPLv3 license. Access to the model's weights was managed by an application process, with access to be granted "on a case-by-case basis to academic researchers; those affiliated with organizations in government, civil society, and academia; and industry research laboratories around the world". Llama was trained on only publicly available information, and was trained at various model sizes, with the intention to make it more accessible to different hardware. The model was exclusively a foundation model, although the paper contained examples of instruction fine-tuned versions of the model. Meta AI reported the 13B parameter model performance on most NLP benchmarks exceeded that of the much larger GPT-3 (with 175B parameters), and the largest 65B model was competitive with state of the art models such as PaLM and Chinchilla.

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Reception Wired describes the 8B parameter version of Llama 3 as being "surprisingly capable" given its size. The response to Meta's integration of Llama into Facebook was mixed, with some users confused after Meta AI told a parental group that it had a child. The release of Llama models has sparked significant debates on the benefits and misuse risks of open-weight models. Such models can be fine-tuned to remove safeguards, notably by cyber criminals, until they comply with harmful requests. Some experts contend that future models may facilitate causing damage more than defending against it, for example by making it relatively easy to engineer advanced bioweapons without specialized knowledge. Conversely, open-weight models can be useful for a wide variety of purposes, including for safety research. Open Source Initiative head Stefano Maffulli criticized Meta for describing Llama as open-source, saying that it was causing confusion among users and "polluting" the term.